

GR 2 Lecture Note

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Abstract

This is the lecture note of the GR 2 course in EPFL 2026 spring semester. Mainly covering the topics in cosmology

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1 Lecture 1: FRW Metric

For a cosmological model, we need to make some assumptions about the universe. The most common assumption is the cosmological principle:

Theorem 1.1

Cosmological Principle

The universe is homogeneous and isotropic on large scales.

The cosmological principle strongly constrains the form of the metric. We assume that we can select a space slice at each time in some coordinate system, and consider the induced metric on each time slice.

1.1 Space Slices Metric

The cosmological principle requires that the metric on each time slice is homogeneous and isotropic, which means that the metric on each time slice is a maximally symmetric space. A maximally symmetric space is a space that has the maximum number of symmetries, which means that the Riemann curvature tensor has the following form:

$$\begin{aligned} R^{(3)} &= \text{constant} \\ R_{ij} &= \frac{1}{3} R^{(3)} g_{ij} \\ R_{ijkl} &= \frac{1}{6} R^{(3)} (g_{ik} g_{jl} - g_{il} g_{jk}) \end{aligned} \tag{1}$$

Thus we can see that a spacial metric can be classified into three types, and in fact each type of metric can be written in a simple form:

- $R^{(3)} = 0$, the metric is $\mathbb{R}^3$ 3-plane
- $R^{(3)} > 0$, the metric is $\mathbb{S}^3$ 3-sphere
- $R^{(3)} < 0$, the metric is $\mathbb{H}^3$ 3-hyperbolic space (Euclidean AdS space)

The metric of $\mathbb{R}^3$ is trivial, we will consider the other two cases in the following:

1.1.1 $\mathbb{S}^3$ 3-sphere

The metric of a 3-sphere can be given by inducing from a 4-dimensional Euclidean space with Cartesian coordinates x^i and the metric $ds^2 = \delta_{ij} dx^i dx^j$. We can induce a coordinate and metric by the following constraint and parametrization:

- Constraint: $\sum_i x_i^2 = a^2$
- Parametrization:

$$\begin{aligned} x_1 &= R \sin \theta \cos \varphi \\ x_2 &= R \sin \theta \sin \varphi \\ x_3 &= R \cos \theta \\ x_4 &= \sqrt{a^2 - R^2} \end{aligned} \tag{2}$$

The induced metric is given by:

$$ds^2 = \frac{a^2 dr^2}{a^2 - r^2} + r^2 d\Omega_2^2 \tag{3}$$

We can calculate the Ricci Scalar of this metric, and we find that $R^{(3)} \sim \frac{1}{a^2}$.

1.1.2 $\mathbb{H}^3$ 3-hyperbolic space

Similarly we can embed the 3-hyperbolic space into a 4-dimensional Minkowski space with Cartesian coordinates x^i to get the metric. Another easy way of seeing the result is simply take the metric of the 3-sphere and replace a^2 with $-a^2$. This will definitely give us a solution to the constraints Equation 1 with negative curvature. Thus we can write the metric of the 3-hyperbolic space as:

$$ds^2 = \frac{a^2 dr^2}{a^2 + r^2} + r^2 d\Omega_2^2 \quad (4)$$

1.1.3 Unified Form of Spacial Metric

With above preparations, we can write three types of metrics in a unified form:

$$ds^2 = a^2 \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_2^2 \right) \quad (5)$$

Here $k = -1, 0, 1$ stands for the three types of metrics. $k = 1$ gives $\mathbb{S}^3$ and $k = -1$ gives $\mathbb{H}^3$.

1.2 FRW Metric

1.2.1 Definition in Comoving Coordinates

With the spacial slice metric prepared, we can add a time coordinate to make it a full 4 dimensional metric ansatz. A typical form of the metric is given by:

Definition 1.2

FRW Metric

The FRW metric is given by:

$$\begin{aligned} ds^2 &= -dt^2 + a^2(t) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega_2^2 \right) \\ &= -dt^2 + a^2(t) \gamma_{ij} dx^i dx^j \end{aligned} \quad (6)$$

Mathematically, we can rigorously show that the FRW metric is the only metric that satisfies the cosmological principle. Physically, we can also understand that the FRW metric is the most general metric that can describe a homogeneous and isotropic universe.

Remark

However, we shall notice that such metric is only defined on a patch of the manifold. Different manifold with different topology can have the same local metric. Thus, the global structure of the spacetime is undetermined.

1.2.2 Unit of the Scale Factor

We can see that:

- For $k \neq 0$;, the scale factor $a(t)$ has the physical meaning of radius of spacial curvature. Thus, we can assign it a unit of length and take the spacial coordinate dimensionless.
- For $k = 0$: the scale factor $a(t)$ can be taken as dimensionless, and the spacial coordinate can be taken as having a unit of length.

In fact in this case, at fixed time, $a(t = t_0)$ is meaningless, what physical is the ratio $\frac{a(t)}{a(t_0)}$.

1.2.3 Remark on Comoving Coordinates

[Definition 1.2](#) is certainly written in a certain coordinate system, which is called the **comoving coordinates**. This means that the particles at rest in this frame (at rest means at rest spacially and moving in the time coordinate) are free (which means they move in geodesics) We can see that indeed $u^0 = 1$ is a solution to the geodesic equation.

1.2.4 Connection and Curvature

We can calculate the Levi-Civita connection and the curvature of the FRW metric in the comoving coordinates:

- Metric:

$$g_{00} = -1, \quad g_{ij} = a^2 \gamma_{ij} \quad g^{00} = -1, \quad g^{ij} = a^{-2} \gamma^{ij} \quad (7)$$

- Curvature:

$$\begin{aligned} R_{00} &= -3 \frac{\ddot{a}}{a} \quad R_{0i} = 0 \\ R_{ij} &= (a\ddot{a} + 2\dot{a}^2 + 2k) \gamma_{ij} \\ R &= 6 \frac{a\ddot{a} + \dot{a}^2 + k}{a^2} \end{aligned} \quad (8)$$

where γ_{ij} is the spacial metric. We can see that the curvature is determined by the scale factor $a(t)$ and the curvature parameter k .

2 Lecture 2: Friedmann Equations and Solutions

We can rewrite the Einstein Field Equations with two ansatz:

- The metric is given by the Friedmann-Robertson-Walker (FRW) metric
- The EM tensor of matter is given by a perfect fluid in the comoving frame

2.1 Einstein Field Equations

We consider the general Einstein Field Equation with a cosmological constant:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi GT_{\mu\nu} \quad (9)$$

2.1.1 Matter Energy-Momentum Tensor in Comoving Frame

In Section 1.2.4 we have calculated the curvature tensor in the comoving frame, which is the LHS of Equation 9. Now we need to calculate the EM tensor of matter in the comoving frame. We focus on:

- The matter is a perfect fluid and being at rest in the comoving frame

which means that the EM tensor has the following form in the comoving frame:

$$T_{\mu\nu} = u_\mu u_\nu (\rho + p) + pg_{\mu\nu} \quad (10)$$

And we assume that the matter is at rest in the comoving frame, which means that the 4-velocity is given by $u^\mu = (1, 0, 0, 0)$. Thus we have the only non-zero component of the EM tensor is:

$$T_{00} = \rho, \quad T_{ij} = pg_{ij} \quad \text{or} \quad T^{00} = \rho, \quad T^{ij} = pg^{ij} = p \frac{\gamma^{ij}}{a^2} \quad (11)$$

Generally, the homogeneity and isotropy of the universe requires that the EM tensor $\rho(t), p(t)$ only depends on time. This suits that the metric is also only depends on time.

Normally, we assume that there is a relation between the energy density and pressure, depend on the matter itself. Thus, the EM tensor has only one degree of freedom, which is the energy density ρ .

2.1.2 Friedmann Equations

Now we can plug in the curvature tensor and the EM tensor into the Einstein Field Equation, and we can get the 00 component of the Einstein Field Equation, which is called the Friedmann Equation:

Theorem 2.1

Friedmann Equation

The Friedmann Equation is given by the 00 component of the EFE of a rest perfect fluid in the comoving frame, which is given by:

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3} \quad (12)$$

Let's now count the variable and equations, usually we fix a k for the model we want to consider. Then, the EM Tensor have 1 DoF $\rho(t)$, the metric have 1 DoF $a(t)$, and we have 1 equation, which is the Friedmann Equation. Thus we need another equation to solve the system.

We first consider the other Terms of the Einstein Field Equation, which is the ij component. We can get another equation from the ij component of the EFE, which is given by:

Theorem 2.2

Spacial Component of EFE

The ij component of the EFE of a rest perfect fluid in the comoving frame is given by:

$$2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} = -8\pi Gp + \Lambda \quad (13)$$

Remark

We only have one spacial equation, this is due to the fact that the spacial metric is dependent on γ_{ij} up to one factor. Thus, the spacial component of the EFE only gives one equation.

Sometimes we combine the two equations together, and we can get rid of the term $\dot{a}$, which is :

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3} \quad (14)$$

From this equation we can easily see when the universe is accelerating or decelerating. If we assume that $\Lambda = 0$ then:

- If $\rho + 3p > 0$ then the universe is decelerating $\ddot{a} < 0$
- If $\rho + 3p < 0$ then the universe is accelerating $\ddot{a} > 0$

2.1.3 Energy Conservation of Matter

Another commonly used choice of the second equation is the energy conservation of matter. Note that this equation is not independent of EFE, since it is given by EFE and the Bianchi Identity.

Theorem 2.3

Energy Conservation of Matter

The energy conservation of matter is given by:

$$\partial_t(\rho a^3) + p\partial_t a^3 = 0 \quad (15)$$

We can see that the energy conservation is equivalent to the first law of thermodynamics, if we view the matter as a thermodynamic system with volume $V = a^3$ and internal energy $U = \rho a^3$.

2.2 Matter Content and Solutions

Now we try to find solutions to the EFE. But we need to specify the matter content exists in the universe.

2.2.1 Matter Content and Equation of State

We usually focus on Model behave as

$$P = \omega\rho \quad (16)$$

And usually the coefficient ω is given by:

- Pressureless Dust: $\omega = 0$
- Relativistic Fluid (Radiation): $\omega = \frac{1}{3}$
- Cosmological Constant: $\omega = -1$

We believe that $\omega \geq -1$ and cc is the lower bound, which is called the null energy condition.

Remark

We can see from the EFE that a matter with $\omega < -1$ gives out $\rho = \text{const}$ in the energy consevation of matter, this is equivalent to having a cosmological constant of $\Lambda = 8\pi G\rho$.

Btw, we may assume that $\rho > 0$ the matter have positive energy density. Thus, the “cosmological constant” case is in fact “positive cosmological constant” case.

2.2.2 Solution 1: Einstein Static Universe

If we assume that $\rho \neq 0$, $p = 0$; $\dot{a} = 0$. which means that the universe is full of pressureless dust and is static. Then the two EFE becomes:

$$\frac{k}{a^2} = \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \frac{k}{a^2} = \Lambda \quad (17)$$

If we take a positive cosmological constant, then we can get the following solution:

$$k = 1 \quad a = \sqrt{\frac{1}{\Lambda}} \quad \rho = \frac{\Lambda}{4\pi G} \quad (18)$$

This solution is called the Einstein Static Universe, which is a static universe with positive curvature and a positive cosmological constant. However, this is unrealistic

2.2.3 Solution 2: Flat Matter Dominated Universe

We assume that $k = 0$, $\Lambda = 0$, and the universe is full of pressureless dust. Then the Friedmann Equation and the Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \partial_t(\rho a^3) = 0 \quad \text{or} \quad \rho a^3 = C \quad (19)$$

we can get a differential equation:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \frac{C}{a^3} \quad (20)$$

The solution to this equation is given by:

$$a(t) = a_0(t - t_0)^{\frac{2}{3}} \quad \rho \sim \frac{1}{t^2} \quad (21)$$

We can notice that this solution $\dot{a} > 0$ and $\ddot{a} < 0$, which means that the universe is expanding and decelerating.

2.2.4 Solution 3: Flat General Matter Dominated Universe

Consider a more general case with $k = 0$, $\Lambda = 0$, and the matter content is given by a general equation of state $P = \omega\rho$. This time the Friedmann Equation and the Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad \dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + p) = 0 \quad (22)$$

This gives us a solution:

$$\rho = Ca^{-3(1+\omega)}, \quad a(t) = a_0(t - t_0)^{\frac{2}{3(1+\omega)}} \quad (23)$$

There are some special properties of this solution:

- **Null Energy Condition** : Notice that we believe that $\omega \geq -1$. Thus we can see that $a(t)$ will always have a singularity at $t = t_0$, for the metric in this coordinate.
- **Matter density time dependent** : We can see that if one kind of matter dominates the universe, then:

$$\rho \sim \frac{1}{t^2} \quad (24)$$

And if we have many kinds of matter and the effective ω is given by ω , then matter ρ_i evolve as:

$$\rho_i \sim t^{-2\frac{1+\omega_i}{1+\omega}} \quad (25)$$

- **Accelerating and Decelerating** : we can calculate $\ddot{a}$ and we see that:
 - If $\omega > -\frac{1}{3}$ then the universe is decelerating $\ddot{a} < 0$
 - If $\omega < -\frac{1}{3}$ then the universe is accelerating $\ddot{a} > 0$
- **Spacial Curvature as Matter**: An interesting observation is that having non-zero spacial curvature k is equivalent as choosing a matter of:

$$\omega = -\frac{1}{3} \quad (26)$$

Thus in fact we can view the spacial curvature as a kind of matter content in the universe. We can also see that this matter makes the universe not accelerating nor decelerating, $\ddot{a} = \text{const.}$

3 Lecture 3: Cosmological Horizons, dS Spacetime and Distance

3.1 Cosmological Horizons

We consider the Penrose diagram of a FRW metric, generally the metric can be written as:

$$ds^2 = -dt^2 + a(t)^2 dx^2 \quad (27)$$

Here we use dx^2 to represent the spacial metric. Then we can make a change of variables:

$$d\eta = \frac{dt}{a(t)} \quad (28)$$

Thus we can have the metric using (η, x) coordinate as:

$$ds^2 = a(\eta)^2 (-d\eta^2 + dx^2) \quad (29)$$

which is a conformally Minkowski metric. Then we need to fix the shape of the Penrose diagram. To do this we have to know what value can we take for η as t goes from $t_0 = 0$ to $t = +\infty$. We can integrate the relation Equation 28 and assume at $t = 0, \eta = \eta_*$:

$$\eta = \int_0^t \frac{dt'}{a(t')} + \eta_* \quad (30)$$

We assume that the space is flat and the spacetime is filled with $P = \omega\rho$ content. Then a general solution is:

$$a(t) \sim t^{\frac{2}{3(1+\omega)}} \quad (31)$$

Then we can see that $\omega > -\frac{1}{3}$ and $\omega < -\frac{1}{3}$ gives two cases for the behavior of η as t goes to zero and infinity.

3.1.1 Case 1 : $\omega > -\frac{1}{3}$ decelerating universe

In the case, we can find that the integral Equation 30 converge at $t = 0$ but diverge at $t = +\infty$. we can choose a suitable η_* to make η go from η_* to $+\infty$, and at $t = 0, \eta = \eta_*$ the metric is singular. Thus the penrose diagram behave as:

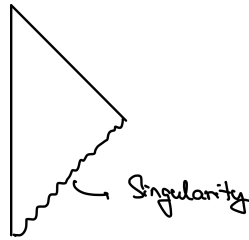


Figure 1: Penrose diagram for a decelerating universe with $\omega > -\frac{1}{3}$

3.1.2 Case 2: $\omega < -\frac{1}{3}$ accelerating universe

In this case, we find the integral Equation 30 diverge at $t = 0$ but converge at $t = +\infty$. This means that for an accelerating universe, we can choose a suitable η_* to make η go from $-\infty$ to a finite number and at $t = 0, \eta = -\infty$ there is a singularity. Thus the penrose diagram be like:

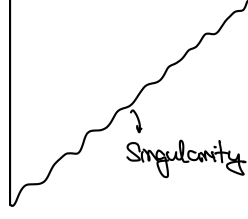


Figure 2: Penrose diagram for an accelerating universe with $\omega < -\frac{1}{3}$

We can see that in this case, there exist something called a **Cosmological Horizon**. It is originate from the fact that two points in the spacetime can never send information to each other, for the light can never reach each other before touching the future infinity.

This is interpret as the Universe is accerlating in expanding, thus the light can never catch up with the expansion of the universe.

3.2 De Sitter Spacetime

3.2.1 Cosmological Constant Case with Flat Spatial Curvature

We now focus on a special case of $k = 0, \Lambda = 0$ universe with the matter content of $\omega = -1$ which is the cosmological constant case. Or equivalently $\Lambda \neq 0, \rho = 0$. Note that it is not include in the general matter case Section 2.2.4, since the solution is distinct.

In this case, the Friedmann Equation and Energy Conservation of Matter becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{\Lambda}{3}, \quad \rho = \text{const} := \frac{\Lambda}{8\pi G} \quad (32)$$

Thus we can get the solution that:

$$a(t) = \exp(Ht), \quad H = \sqrt{\frac{\Lambda}{3}}, \quad (33)$$

This solution we can see that t goes from $-\infty$ to $+\infty$. There is a singularity at $t = -\infty$ where $a(t) = 0$.

Remark

Here we take a convention that $a_0 = 1$ and in fact there are two solutions e^{Ht} and e^{-Ht} . We can choose the solution e^{Ht} since it is more physical, which means that the universe is expanding. The other solution e^{-Ht} is a contracting universe, and singularity at $t = +\infty$ where $a(t) = 0$. However, this solution is not physical since we believe that the universe is expanding, thus we can discard this solution.

3.2.2 Penrose Diagram

We can make a change of variable:

$$\eta = \int \frac{dt}{a(t)} = -\frac{e^{-Ht}}{H} \quad (34)$$

Thus we can get the metric in the (η, x) then the metric becomes:

$$ds^2 = \frac{a_0^2}{H^2 \eta^2} (-d\eta^2 + dx^2) \quad (35)$$

we can see the range of the parameter η :

- as $t \rightarrow \infty$ $\eta \rightarrow 0^-$ which is far future.
- as $t \rightarrow -\infty$ $\eta \rightarrow -\infty$ which is the past singularity.

Thus the penrose diagram of this spacetime is like:

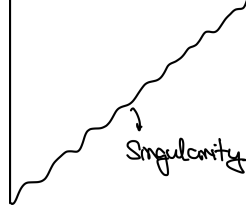


Figure 3: Penrose diagram for cosmological constant case with $\omega < -1$

which behaves just like the accerlating universe case in the general matter solution Figure 2.

3.2.3 Global de Sitter Spacetime

However, the singularity we get at $t = -\infty$ is not a real singularity!! These can be seen by the behavior of the curvature. We can prove that the curvature is a constant and don't have any singularity at $t = -\infty$. Thus, we expect we can do an extension of the spacetime to get rid of this coordinate singularity.

One approach to this is to investigate a seemingly different solution of the EFE, which is the **vaccum solution with $k = 1$** . The EFE reads:

$$\left(\frac{\dot{a}}{a}\right)^2 + \frac{1}{a^2} = \frac{\Lambda}{3}, \quad 2\left(\frac{\ddot{a}}{a}\right) + \left(\frac{\dot{a}}{a}\right)^2 + \frac{1}{a^2} = \Lambda \quad (36)$$

The solution to this equation is given by:

$$a(t) = H^{-1} \cosh(Ht), \quad H = \sqrt{\frac{\Lambda}{3}}, \quad (37)$$

This spacetime is called **Global dS Spacetime**.

$$ds^2 = -dt^2 + H^{-2} \cosh^2(Ht)(d\chi^2 + \sin^2(\chi)d\Omega_2^2) \quad (38)$$

We can see that the metric is non-singular at any t for the cosh function is well behaved and have no zero point. More importantly, we can prove:

- The cosmological constant case with flat spatial curvature is a patch of the global dS spacetime, which is called the **Poincare Patch**.

In penrose diagram, this can be shown as:

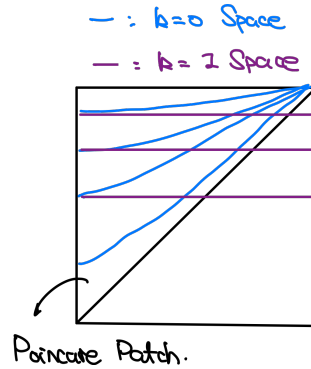


Figure 4: Penrose diagram for the global dS spacetime, where the blue region is the Poincare patch which is the cosmological constant case with flat spatial curvature

We can see that the blue line is a spacial slicing that make the spacial metric flat, yet the price to pay is that it only cover a patch of the global dS spacetime.

3.3 Hubble Parameter

We recap on the three equations we commonly use for solving EFE:

$$\begin{aligned} \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} &= \frac{8\pi G}{3}\rho + \frac{\Lambda}{3}, \quad \text{Friedmann Euqation} \\ 2\left(\frac{\ddot{a}}{a}\right) + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} &= -8\pi Gp + \Lambda, \quad \text{Spacial EFE} \\ \dot{\rho} + 3\left(\frac{\dot{a}}{a}\right)(\rho + p) &= 0 \quad \text{Energy Conservation of Matter} \end{aligned} \tag{39}$$

If we consider a certain kind of matter $p_i = \omega_i \rho_i$, the enegy conservation gives us:

$$\rho_i = C a^{-3(1+\omega_i)}, \quad p_i = \omega_i \rho_i \tag{40}$$

Moreover, we can define a parameter:

Definition 3.1

Hubble Parameter

The **Hubble Parameter** is defined as:

$$H = \frac{\dot{a}}{a} \tag{41}$$

Thus the Friedmann Equation with $k = 0, \Lambda = 0$ can be written as:

$$H^2 = \frac{8\pi G}{3} \sum_i \rho_i \tag{42}$$

Further more we can define a parameter called **Critical Density** as:

Definition 3.2

Critical Density

The **Critical Density** is defined as:

$$\rho_c = \frac{3H_0^2}{8\pi G}, \quad \text{equivalently} \quad \rho_c = \sum_i \rho_i(t = t_0) \tag{43}$$

where H_0 is the Hubble Parameter at present time t_0 . And we can define Ω_i which is :

$$\Omega_i = \frac{\rho_i(t = t_0)}{\rho_c}, \quad \text{thus} \quad \sum_i \Omega_i = 1 \tag{44}$$

With all these preparation, we can rewrite the Friedmann Equation (with $k \neq 0, \Lambda \neq 0$ and with the energy conservation already used) as:

$$H^2 = H_0^2 \left(\Omega_\Lambda + \Omega_k \left(\frac{a_0}{a} \right)^2 + \Omega_m \left(\frac{a_0}{a} \right)^3 + \Omega_\gamma \left(\frac{a_0}{a} \right)^4 \right) \tag{45}$$

The first two term is the contribution from the cosmological constant and the spatial curvature, and the last two term is the contribution from the matter and the radiation, with certain ω_i determine its behavior.

If we rewrite LHS back with $\frac{\dot{a}}{a}$ and we group terms we can get:

$$\dot{a}^2 = H_0^2 \left(\Omega_\Lambda a^2 + \Omega_k a_0^2 + \Omega_m \frac{a_0^3}{a} + \Omega_\gamma \frac{a_0^4}{a^2} \right) \quad (46)$$

Which can be interpreted as the equation of motion of a particle with zero total energy moving in a potential:

$$U(a) = -H_0^2 \left(\Omega_\Lambda a^2 + \Omega_k a_0^2 + \Omega_m \frac{a_0^3}{a} + \Omega_\gamma \frac{a_0^4}{a^2} \right), \quad \dot{a}^2 + U(a) = 0 \quad (47)$$

3.4 Distances in Cosmology

To investigate the cosmology, we need to define the lengths.

3.4.1 Redshift

Assume a process, a star emit a light at $t = t_1, r = r_1$ and the light is observed by us at $t = t_0, r = 0$. The light travel along null geodesics:

$$ds^2 = -dt^2 + a(t)^2 \frac{dr^2}{1 - kr^2} = 0 \quad \Rightarrow \quad dt = -a(t) \frac{dr}{\sqrt{1 - kr^2}} \quad (48)$$

There is a “-” sign since the light is traveling from r_1 to 0. Thus we can get evaluate a length of the light travel $l(r)$:

$$\int_{t_1}^{t_0} \frac{dt}{a(t)} = \int_0^{r_1} \frac{dr}{\sqrt{1 - kr^2}} = l(r_1) \quad (49)$$

We can calculate the length $l(r)$ for different spatial curvature:

$$l(r) = \begin{cases} \arcsin(r) & k = 1 \\ r & k = 0 \\ \operatorname{arcsinh}(r) & k = -1 \end{cases} \quad (50)$$

Then we can consider a process of two light signal emit at $t = t_1$ and $t = t_1 + \delta t_1$ and the light is observed at $t = t_0$ and $t = t_0 + \delta t_0$. Since the two light signal travel along the same null geodesics, we can get:

$$\frac{\delta t_0}{a(t_0)} = \frac{\delta t_1}{a(t_1)} \quad (51)$$

The frequency of the light is inversely proportional to the time interval $\omega \propto \frac{1}{t}$, thus we can get:

$$\frac{\omega_0}{\omega_1} = \frac{\delta t_1}{\delta t_0} = \frac{a(t_0)}{a(t_1)} \quad (52)$$

- Universe is Expanding: $a(t_0) > a(t_1)$ thus $\omega_0 > \omega_1$ which is called **Redshift**.
- Universe is Contracting: $a(t_0) < a(t_1)$ thus $\omega_0 < \omega_1$ which is called **Blueshift**.

The redshift can be described by the following quantity:

Definition 3.3

The **Redshift** is defined as:

$$z = \frac{\lambda_0 - \lambda_1}{\lambda_0} = \frac{a(t_0)}{a(t_1)} - 1 \quad (53)$$

note that t_0 is the time of getting the light and t_1 is the time of emitting the light.

Redshift

Usually, we know the λ_1 of many processes, and we measure λ_0 from experiments. Thus, we can use z as a measure of the distance.

3.4.2 Physical Distance and Coordinate Distance

For two comoving objects at $r = r_1$ and $r = r_2$, the coordinate distance is:

$$l_c = l(r_1) - l(r_2) \quad (54)$$

which is the distance separated in the spacial coordinate. However, its not the distance in the general metric. The physical distance is given by:

$$l_p = a(t)(l(r_1) - l(r_2)) = a(t)l_c \quad (55)$$

We know that for a comoving object, the spacial coordinate is fixed when moving in geodesics. Yet, the physical distance is time dependent due to the $a(t)$ parameter:

$$\frac{dl_p}{dt} = \dot{a}l_c = Hl_p, \quad H = \frac{\dot{a}}{a} \quad (56)$$

This is the **Hubble Law**, which fits well with the observation of the universe, which is one of the evidences for the expanding universe.

Remark

We here have three notion of “length”:

- r , the spacial coordinate variable, which is NOT the coordinate distance.
- $l(r)$, the coordinate distance, which is the distance in the spacial metric.
- l_p , the physical distance, which is the distance in the full metric.

3.4.3 Luminosity Distance

We first define something as:

- **Standard Candle** : which is a object with known luminosity L , where $L = \frac{dE}{dt}$ is defined as the energy emitted by the object per unit time in the object’s rest frame.

Then we consider a comoving telescope observing the standard candle, we can define a quantity called **Luminosity Distance** as:

Definition 3.4

Luminosity Distance

The **Luminosity Distance** is defined as:

$$P = L \frac{S}{4\pi d^2} \quad (57)$$

where P is the measured energy recieved per unit time by the telescope, and S is the area of the telescope. The luminosity distance d is defined as the distance that make the above equation hold.

Then we can calculate the luminosity distance and its relation with other quantities for a comoving telescope and star in the FRW metric set up.

In the FRW metric set up, the actual P can be calculated as:

$$P = L \left(\frac{S}{S_{\text{tot}}} \right) \left(\frac{\hbar\omega_0}{\hbar\omega_1} \right) \left(\frac{\delta t_1}{\delta t_0} \right) \quad (58)$$

Lets explain each term, we set convension of t_1 the time of emitting the light and t_0 the time of getting the light:

- L is the Energy emitted per unit time when the energy is emitted.
- S is the area of the telescope. S_{tot} is the total area of light front, due to the expansion of the universe it takes form of:

$$S_{\text{tot}} = 4\pi a(t_0)^2 r^2(t_1, t_0) \quad (59)$$

where $r(t_1, t_0)$ is the difference in coordinate r between the star and the telescope (which is NOT the coordinate distance), which follow the null geodesic equation as:

$$dt = -a(t) \frac{dr}{\sqrt{1 - kr^2}} \Rightarrow \int_{t_1}^{t_0} \frac{dt}{a(t)} = \int_0^r \frac{dr}{\sqrt{1 - kr^2}} \quad (60)$$

where r then is implicitly a function of t_1 and t_0 .

Remark

Why do this S_{tot} is the total area of the light front?

We can see this from the FRW metric [Definition 1.2](#), for a fixed time t_0 , and a fixed r , the area of the sphere is given by:

$$ds^2 = a^2(t_0) r^2 d\Omega_2^2 \quad (61)$$

we integrate over the angular coodiantes we can get the area of the sphere as:

$$S = 4\pi a^2(t_0) r^2 \quad (62)$$

- $\frac{\omega_0}{\omega_1}$ is the ratio of the frequency of the light when getting and emitting, which is given by the redshift as:

$$\frac{\omega_0}{\omega_1} = \frac{a(t_0)}{a(t_1)} \quad (63)$$

- $\frac{\delta t_1}{\delta t_0}$ is the time given by the difference of infinitesimal time changes. This is also given by the redshift, see Equation 51:

$$\frac{\delta t_1}{\delta t_0} = \frac{a(t_1)}{a(t_0)} \quad (64)$$

If we put all these together, we can get:

$$P = L \left(\frac{S}{S_{\text{tot}}} \right) \left(\frac{\hbar \omega_0}{\hbar \omega_1} \right) \left(\frac{\delta t_1}{\delta t_0} \right) = L \left(\frac{S}{4\pi a(t_0)^2 r^2} \right) \left(\frac{a(t_0)}{a(t_1)} \right) \left(\frac{a(t_1)}{a(t_0)} \right) = L \frac{S}{4\pi a(t_0)^2 r^2} \quad (65)$$

This gives us the luminosity distance as:

Corollary 3.5

Luminoity Distance in FRW Metric

The luminosity distance in the FRW metric is given by:

$$d = \frac{a(t_0)^2}{a(t_1)} r(t_1, t_0) = (1 + z) a(t_0) r(t_1, t_0) \quad (66)$$

where z is the redshift, see [Definition 3.3](#).

4 Lecture 4: Thermal History of the Universe I: Equilibrium

4.1 Distances in Cosmology II

4.1.1 Luminosity Distance and Redshift

We want a formula for the luminosity distance d_L as a function of redshift z . Thus, if we both measure d_L and z then we can find measure parameter $a(t)$.

We start from the expression [Corollary 3.5](#), we can see that $r(t_0, t_1)$ can be calculated using the null geodesic Equation 60, using the result of Equation 50 gives:

$$r(t_0, t_1) = \begin{cases} \int_{t_1}^{t_0} \left(\frac{dt}{a(t)} \right) & k = 0 \\ \sin \left(\int_{t_1}^{t_0} \left(\frac{dt}{a(t)} \right) \right) & k = 1 \\ \sinh \left(\int_{t_1}^{t_0} \left(\frac{dt}{a(t)} \right) \right) & k = -1 \end{cases} \quad (67)$$

Then we want to calculate the integral:

$$\int_{t_1}^{t_0} \left(\frac{dt}{a(t)} \right) \quad (68)$$

For a general universe, the Friedmann equation and Energy conservation equation gives Equation 45. We copy it here:

$$H^2 = H_0^2 \left(\Omega_\Lambda + \Omega_k \left(\frac{a_0}{a} \right)^2 + \Omega_m \left(\frac{a_0}{a} \right)^3 + \Omega_\gamma \left(\frac{a_0}{a} \right)^4 \right) \quad (69)$$

Remark

Note that in this equation we set a time t_0 to define H_0 , a_0 and Ω_i these constants. Here in the context of distance, we take t_0 to be the time when the light is emitted (note this conflicted with the previous convention where we take t_0 to be the time when the light is observed, and t_1 to be the time when the light is emitted).

We make a change of variable $x = \frac{a}{a_0}$, then we have:

$$\dot{x} = H_0 A(x) x, \quad A(x)^2 = (\Omega_\Lambda + \Omega_k x^{-2} + \Omega_m x^{-3} + \Omega_\gamma x^{-4}) \quad (70)$$

With these preparation we can calculate the integral:

$$\int_{t_1}^{t_0} \left(\frac{dt}{a(t)} \right) = \frac{1}{a_0 H_0} \int_{x_1}^{x_0} \left(\frac{dx}{A(x) x^2} \right) \quad x_0 = 1, x_1 = \frac{a(t_1)}{a_0} = \frac{1}{1+z} \quad (71)$$

Then the final formula for the luminosity distance is given by:

Corollary 4.1

Luminosity Distance to Redshift

The luminosity distance can be calculated as a function of redshift as:

$$d(z) = a_0(1+z) \frac{1}{\sqrt{-k}} \sinh \left(\frac{\sqrt{-k}}{a_0 H_0} \int_{\frac{1}{1+z}}^1 \left(\frac{dx}{A(x)x^2} \right) \right) \quad (72)$$

This is combining the three cases of k into one formula, for $k = 0$ we take the limit $k \rightarrow 0$ and use the fact that $\lim_{x \rightarrow 0} \left(\frac{\sinh(x)}{x} \right) = 1$.

4.1.2 Parallax

Consider we have a source very far from the sun and we want to measure the distance between another star, which is much closer to the sun (though still very far from us). We can measure the angle between the source and the star at two different time (for example, at two different position of the earth in its orbit around the sun). This is called parallax.

See the diagram for illustration:

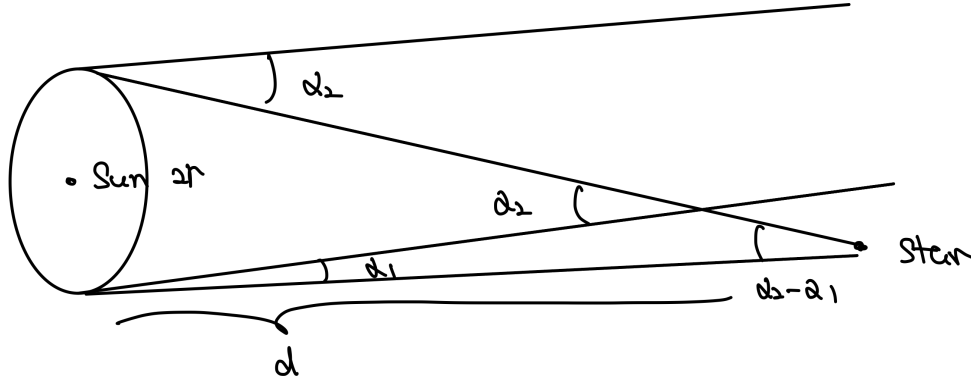


Figure 5: Parallax measurement.

The parallax is defined as:

Definition 4.2

Parallax

The parallax is defined as the angle between the source and the star at two different time.

$$\alpha_2 - \alpha_1 \quad (73)$$

A distance can be calculated from the parallax as:

$$d = \frac{2r}{\alpha_2 - \alpha_1} \quad (74)$$

where r is the distance between the sun and the earth.

4.2 Review on Euquilibrium Statistical Mechanics

4.2.1 Occupation Number and Thermal Distribution Function

For a Equilibrium system, Bosons and Fermions have different occupational number:

Theorem 4.3**Occupation Number of Bosons and Fermions**

For a system in equilibrium, the occupation number of bosons and fermions are given by:

$$n_B(E) = \frac{1}{e^{\frac{E-\mu}{T}} - 1}, \quad n_F(E) = \frac{1}{e^{\frac{E-\mu}{T}} + 1} \quad (75)$$

where E is the energy of the state, μ is the chemical potential and T is the temperature.

The thermal distribution function of the particles are given by $f(p)$, it is related to the occupation number and the number of dof of the particle:

Definition 4.4**Thermal Distribution Function**

The thermal distribution function of the particle i is given by:

$$f_i(p) = \left(\frac{g_i}{(2\pi)^3} \right) n_i(E(p)) \quad E(p) = \sqrt{p^2 + m_i^2} \quad (76)$$

Remark

As a reminder of convention of statistical mechanics. the $(2\pi\hbar)^3$ or in the above formula $(2\pi)^3$ where we set $\hbar = 1$ is the density of phase space. and g_i is the number of degrees of freedom of the particle, for example, for a photon we have $g_i = 2$ because it has two polarization states.

For total number of particle, we need to integrate over the phase space $d^3x d^3p$, yet we leave the spacial part d^3x out because we are interested in the number density, which is the number of particle per unit volume. Thus we only integrate over the momentum space d^3p .

Integrating over $f_i(p)$ gives the **number density** of the particle and integrating with the weight of energy gives the **energy density** of the particle:

- **Number Density:**

$$N_i = \int f_i(p) d^3p \quad (77)$$

- **Energy Density:**

$$\rho_i = \int E(p) f_i(p) d^3p \quad (78)$$

4.2.2 Energy and Number Density in Relativistic Limit

The number density and energy density at equilibrium can be calculated explicitly. For particles in the **relativistic limit**, we means really high temperature::

$$T \gg m_i \Rightarrow T \sim E \sim |p| \quad (79)$$

we can calculate the number density and energy density as:

- **Energy Density:**

$$\rho_i = \begin{cases} \left(\frac{\pi^2}{30} \right) g_i T^4 & \text{Boson} \\ \frac{7}{8} \left(\frac{\pi^2}{20} \right) g_i T^4 & \text{Fermion} \end{cases} \quad (80)$$

- **Number Density:**

$$N_i = \begin{cases} \left(\frac{\zeta(3)}{\pi^2}\right) g_i T^3 & \text{Boson} \\ \frac{3}{4} \left(\frac{\zeta(3)}{\pi^2}\right) g_i T^3 & \text{Fermion} \end{cases} \quad (81)$$

We can also calculate the average energy of the particle in this limit:

$$\langle E \rangle = \frac{\rho_i}{N_i} \quad (82)$$

For the convenience of calculating energy density of system with multiple particle species, we can define the **effective number of species**:

$$g_* = \sum_{\text{boson}} g_i + \left(\frac{7}{8}\right) \sum_{\text{fermion}} g_i \quad (83)$$

and the energy density of the system can be calculated as:

$$\rho = \left(\frac{\pi^2}{30}\right) g_* T^4 \quad (84)$$

4.2.3 Entropy Density in Relativistic Limit

From the second law of thermodynamics, we have:

$$E = TS - PV + \mu N \quad (85)$$

we take the assumption that the chemical potential is zero, then we have:

$$s = \frac{S}{V} = \frac{\rho + P}{T} \quad (86)$$

where ρ is the energy density and P is the pressure of the system. For a relativistic particle remeber we have Equation 16, we take $P = \frac{1}{3}\rho$ for relativistic particle, thus we have:

$$s = \frac{4}{3} \left(\frac{\rho}{T}\right) \quad (87)$$

4.2.4 Energy and Number Density in Non-Relativistic Limit

For non-relativistic limit, we means that:

$$m_i \gg T \Rightarrow E \sim m_i + \frac{p^2}{2m_i} \quad (88)$$

Note that here μ can be important. In this region the number density and energy density can be calculated as:

- **Number Density:**

$$N_i = g_i \left(\frac{m_i T}{2\pi}\right)^{\frac{3}{2}} \exp\left(\frac{\mu_i - m_i}{T}\right) \quad (89)$$

- **Energy Density:**

$$\rho_i = g_i m_i N_i(T) \quad (90)$$

Moreover the pressure of the particle is given by:

$$P_i = g_i N_i T \ll \rho_i \quad (91)$$

Thus, we can see Equation 16 that the pressureless dust approximation is valid for non-relativistic particle.

4.3 CMB Temperature

With these preparation of thermodynamics, we can calculate the CMB temperature. We first assume that the universe is:

- $k = 0, \Lambda = 0$ and dominated by relativistic particles

then the Friedmann equation gives Equation 42 :

$$H = \left(\frac{8\pi G}{3} \rho_{\text{rad}} \right)^{\frac{1}{2}} \quad (92)$$

And from the thermodynamics calculation we have Equation 84, we then can see the temperature dependency of the Hubble parameter:

$$H = \frac{T^2}{M_0} \quad M_0 = \sqrt{\frac{45}{4\pi^3 G g_*}} \quad (93)$$

where g_* is the effective number of species at the time.

For a radiation dominated universe, we have $a(t) \sim t^{\frac{1}{2}}$, or if we choose a time t_0 to define a_0 we have $a(t) = a_0 \left(\frac{t}{t_0} \right)^{\frac{1}{2}}$. Then:

$$H = \frac{\dot{a}}{a} = \frac{1}{2t} \quad (94)$$

Finally, we arrive at the relation between temperature and time:

$$\frac{T^2}{M_0} = \frac{1}{2t} \quad \Rightarrow \quad T = \sqrt{\frac{M_0}{2t}} \quad (95)$$

Now we can estimate with real numbers. g_* we can plug in the standard model particles which gives $g_* \sim 100$, and time of today's universe.

5 Lecture 5: Thermal History of the Universe II: Non-Equilibrium

The Universe is not always in thermal equilibrium, now we have to consider the non-equilibrium processes in the universe.

5.1 Boltzmann Equation

5.1.1 A Convention Shift

For a non-equilibrium system, we need to consider the Boltzmann equation, which describes the evolution of the distribution function of particles in phase space. Here we use a different convention (as in the lecture note), we have:

$$n_i(p, t) = (2\pi)^3 f_i(p, t) = g_i n_{i \text{ old}}(p, t) \quad (96)$$

Then the number density is:

$$N_i = \int \frac{d^3p}{(2\pi)^3} n_i(p, t) \quad (97)$$

and similarly for the energy density.

An important remark is that the distribution p we use the **physial momentum**, for detailed definition see next subsection.

5.1.2 Free Particle in an Expanding Universe

We first consider the free particle, in an expanding universe. The Lagrangian is given by:

$$\mathcal{L} = \frac{1}{2} m g_{ij}(\dot{x}^i \dot{x}^j) \quad (98)$$

here $\dot{x}$ is the derivative with respect to the coordinate time t . We now can explicitly define two momentums:

- **Physical Momentum:** $\mathcal{L} = \frac{p_p^2}{2m}$ this gives us: $p_p^i = m a(t) \dot{x}^i$.
- **Comoving Momentum:** $p_c^i = \frac{\partial \mathcal{L}}{\partial \dot{x}^i}$ this gives us: $p_c^i = m a(t)^2 \dot{x}^i$.

Then from the Euler-Lagrange equation, we can get the equation of motion of the particle, which is given by:

$$\frac{d}{dt} p_c^i = 0 \quad (99)$$

Remark

This can also be view given by the Noether's theorem of spacial translation symmetry.

We then can see that the comoving momentum is conserved, which means that the physical momentum is redshifted by the expansion of the universe:

$$p_p^i = \frac{p_c^i}{a(t)} = \frac{\text{constant}}{a(t)} \quad (100)$$

if we choose an “initial time” t_0 and define the physical momentum at that time as p_0 and $a(t_0) = a_0$, then we can write the physical momentum at any time as:

$$p_0^i = \frac{p_c^i}{a_0} = p_p^i \frac{a(t)}{a_0} \quad (101)$$

Thus given a time t , we want to measure the number of particle with physical momentum p , due to the fact that the comoving momentum is conserved, the number is equal to number of particle at t_0 with physical momentum $p \frac{a(t)}{a_0}$. Thus we have:

$$n_i(p, t) = n_i\left(p \frac{a(t)}{a_0}, t_0\right) \equiv n_{0i}\left(p \frac{a(t)}{a_0}\right) \quad (102)$$

5.1.3 Free Particle Boltzmann Equation

Then we can derive the Boltzmann equation for a free particle in an expanding universe. We have:

$$\frac{\partial}{\partial t} n_i(p, t) = \frac{d}{dt} n_{0i}\left(p \frac{a(t)}{a_0}\right) = (n'_{0i}) \times \left(p \frac{\dot{a}(t)}{a_0}\right) \quad (103)$$

and

$$\frac{\partial}{\partial p} n_i(p, t) = n'_{0i} \times \left(\frac{a(t)}{a_0}\right) \quad (104)$$

we combine the two equations together, we can get:

$$\frac{\partial}{\partial t} n_i(p, t) - \left(\frac{\dot{a}(t)}{a(t)} p\right) \frac{\partial}{\partial p} n_i(p, t) = 0 \quad (105)$$

Then we have the Boltzmann equation for a free particle in an expanding universe:

Theorem 5.1

Free Particle Boltzmann Equation

The Boltzmann equation for a free particle in an expanding universe is given by:

$$\frac{\partial}{\partial t} n_i(p, t) - (Hp) \frac{\partial}{\partial p} n_i(p, t) = 0 \quad (106)$$

where $H = \frac{\dot{a}(t)}{a(t)}$ is the Hubble parameter.

We can see that a “free particle” in an expanding universe is not really free, it is affected by the expansion of the universe, which causes the redshift of the physical momentum.

We then can integrate the Boltzmann equation over the momentum space, we can get the equation for the number density:

$$\dot{N}_i(t) + 3H(t)N_i(t) = 0 \quad (107)$$

Remark

Remember the convention is changed and we integrate with $\frac{d^3p}{(2\pi)^3}$ measure.

We'd mainly prefer using number density instead of distribution function as the variable.

5.1.4 Interacting Particle Boltzmann Equation

Then we consider a particles with interactions. The generalization is done by adding a collision term I_{col} to the Boltzmann equation:

Theorem 5.2**Interacting Particle Boltzmann Equation**

The Boltzmann equation for an interacting particle in an expanding universe is given by:

$$\frac{\partial}{\partial t} n_i(p, t) - (Hp) \frac{\partial}{\partial p} n_i(p, t) = I_{\text{col}} \quad (108)$$

where I_{col} is the collision term, which describes the interactions of the particle with other particles.

If we focus on **2 to 2 scattering procedure**, the collision term can be written in terms of the scattering amplitude of the procedure, which is given by:

$$I_{\text{col}} = -\frac{1}{2E_p} \int \frac{d^3 q_1}{(2\pi)^3} \frac{d^3 q_2}{(2\pi)^3} \frac{d^3 q_3}{(2\pi)^3} (2\pi)^4 \delta^{(4)}(p + q_1 - q_2 - q_3) |\mathcal{M}|^2 \times [n(p)n(q_1)(1 \pm n(q_2))(1 \pm n(q_3)) - n(q_3)n(q_2)(1 \pm n(p))(1 \pm n(q_1))] \quad (109)$$

This equation is quite impossible to solve analytically, yet we have the technique of **Relaxation Time Approximation**, which we expand around the equilibrium distribution $n_i^{\text{eq}}(p)$.

5.1.5 Relaxation Time Approximation

We consider at $H = 0$ we have the equilibrium distribution $n_i^{\text{eq}}(p)$, we can prove that:

$$I_{\text{col}}(n_i^{\text{eq}}(p)) = 0 \Leftrightarrow \frac{d}{dt} n_i^{\text{eq}}(p) = 0 \quad (110)$$

where n_i^{eq} is exactly the Bose-Einstein distribution or the Fermi-Dirac distribution. Then we can expand the distribution around the equilibrium distribution, in this limit:

$$I_{\text{col}}(n_i) = I_{\text{col}}(n_i^{\text{eq}} + \delta n_i) = I_{\text{col}}(n_i^{\text{eq}}) + \left(\frac{\delta I_{\text{col}}}{\delta n_i} \right) \delta n_i + \dots = -\Gamma \times (n_i - n_i^{\text{eq}}) \quad (111)$$

Thus the Boltzmann equation becomes:

$$\frac{\partial}{\partial t} n_i(p, t) - (Hp) \frac{\partial}{\partial p} n_i(p, t) = -\Gamma \times (n_i - n_i^{\text{eq}}) \quad (112)$$

or if we integrate over the momentum space, we can get the equation for the number density:

Theorem 5.3**Relaxation Time Approximation Boltzmann Equation**

The Boltzmann equation for an interacting particle in an expanding universe in the relaxation time approximation is given by:

$$\dot{N}_i(t) + 3H(t)N_i(t) = -\Gamma \times (N_i(t) - N_i^{\text{eq}}) \quad (113)$$

Now we talk more about Γ . It has a physical meaning as the **rate of reaction**:

$$\Gamma = \frac{1}{\tau} = \left\langle \frac{\lambda}{v} \right\rangle^{-1} \quad (114)$$

where τ is the **mean free time** of the collision and λ is the **mean free path** of the collision and v is the **relative velocity** of the particle. The mean free path is related to the **cross section** of the collision by:

$$\lambda = \frac{1}{\sigma N} \quad (115)$$

where σ is the cross section of the collision and N is the number density of the particle. Thus we can see that Γ is not time independent, it depends on the number density of the particle and the velocity of particle, which can change with time.

5.2 Freeze-in and Freeze-out

With this Boltzmann equation, we can now understand the process of **freeze-in** and **freeze-out** of particles in the early universe. Note that from the Boltzman equation, the number of particles changes due to two effects:

- **Expansion of the universe:** which is given by the term $3H(t)N_i(t)$.
- **Interactions of the particle:** which is given by the term $-\Gamma \times (N_i(t) - N_i^{\text{eq}})$.

Both terms are given by a rate Γ and H . We can find that both rates are **Temperature Dependent**:

- Γ depends on v the relative velocity of the particle, which is related to the temperature of the universe.
- H depends on time as univers expands, which is also related to the temperature of the universe Equation 95.

Thus, as the universe cools down, different terms will dominate at different temperature, which gives us the process of freeze-in and freeze-out.

5.2.1 Asymptotic Behavior of the Boltzmann Equation

We discuss first the asymptotic solution of the Boltzmann equation. We have two limits:

- $\Gamma(T) \gg H(T)$: in this limit, the interaction term dominates, thus we have

$$N_i(t) \approx N_i^{\text{eq}} \quad (116)$$

which means that the particle is in thermal equilibrium with the universe.

- $\Gamma(T) \ll H(T)$: in this limit, the expansion term dominates, Boltzmann equation becomes:

$$\frac{d}{dt}(a^3 N_i) = 0 \quad (117)$$

Thus, the number density deviate from the equilibrium value, and we have:

$$N(T) \approx N^{\text{eq}}(T_*) \left(\frac{a(T_*)}{a(T)} \right)^3 \quad (118)$$

where T_* is the temperature at which $\Gamma(T_*) \approx H(T_*)$. This means the time we should not only consider the expansion of the universe, but also the interaction of the particle.

The critical temperature T_* is called the **freeze-out/in temperature** of the particle, depends on $\frac{\Gamma}{H}$ is decreasing or increasing with temperature, we can have freeze-out or freeze-in process.

5.3 Electron-Positron Gas Freeze-out

We finally can come to a concrete example of the freeze-out process, which is the freeze-out of the electron-positron gas in the early universe. We consider the following scattering process:

$$e^+ e^- \leftrightarrow \gamma \gamma \quad (119)$$

With this process occuring we assume that the universe is **radiation dominant** !!

5.3.1 Scattering Cross Section and Rate of Reaction

The scattering cross section of this process is given by:

$$\sigma = \begin{cases} \frac{1}{2v} \pi r_e^2 & v \ll 1 \\ \frac{m_e^2}{E^2} \pi r_e^2 \left(\log \left(\frac{(2E)^2}{m_e^2} \right) - 1 \right) & v \approx 1 \end{cases} \quad (120)$$

where r_e is $\frac{\alpha}{m_e}$, E is the center of mass energy of the collision, and v is the relative velocity of the particle.

This cross section is depending on the velocity of the particle, which is related to the temperature of the universe. In both limits we can calculate the rate of reaction Γ :

- **Relativistic Limit:** $T \geq m_e$ in this limit we take:

$$\sigma \approx \frac{m_e^2}{E^2} \pi r_e^2 \left(\log \left(\frac{(2E)^2}{m_e^2} \right) - 1 \right) \approx \frac{\alpha^2}{T^2} \quad (121)$$

Remark

In this approximation we take $E \approx T$ for relativistic particle and we assume the logarithm term is order of 1.

and remember that the number density of relativistic particle for equilibrium case is given by:

$$N_i^{\text{eq}} = \frac{3}{4} \left(\frac{\zeta(3)}{\pi^2} \right) g_i T^3 \quad (122)$$

Thus we have:

$$\Gamma = \langle N v \sigma \rangle \sim \alpha^2 T \quad (123)$$

- **Non-Relativistic Limit:** $T \leq m_e$ in this limit we take:

$$\sigma \approx \frac{1}{2v} \pi r_e^2 \quad (124)$$

and remember that the number density of non-relativistic particle for equilibrium case is given by:

$$N_i^{\text{eq}} = g_i \left(\frac{m_i T}{2\pi} \right)^{\frac{3}{2}} \exp \left(\frac{\mu_i - m_i}{T} \right) \quad (125)$$

Thus we have:

$$\Gamma = \langle N v \sigma \rangle \sim r_e^2 (T m_e)^{\frac{3}{2}} \exp \left(\frac{-m_e}{T} \right) \quad (126)$$

Remark

We here take the chemical potential μ_i to be zero, which is a good approximation in the radiation dominant universe.

5.3.2 Freeze-in/out Temperature

We list out the rate of reaction and the Hubble parameter:

$$\Gamma = \begin{cases} \sim \alpha^2 T & T \geq m_e \\ \sim r_e^2 (T m_e)^{\frac{3}{2}} \exp\left(\frac{-m_e}{T}\right) & T \leq m_e \end{cases} \quad (127)$$

while the Hubble parameter (in a radiation dominant univers) is given by Equation 93, we copy it here:

$$H = \frac{T^2}{M_0} \quad (128)$$

Remark

We here are assuming that the universe happening all those processes is radiation dominant, which is a good approximation for the early universe. Thus, we can negelact the chemical potential of the particle, and use this Hubble parameter to calculate the freeze-out/in temperature.

We draw the plot of Γ and H as a function of temperature:

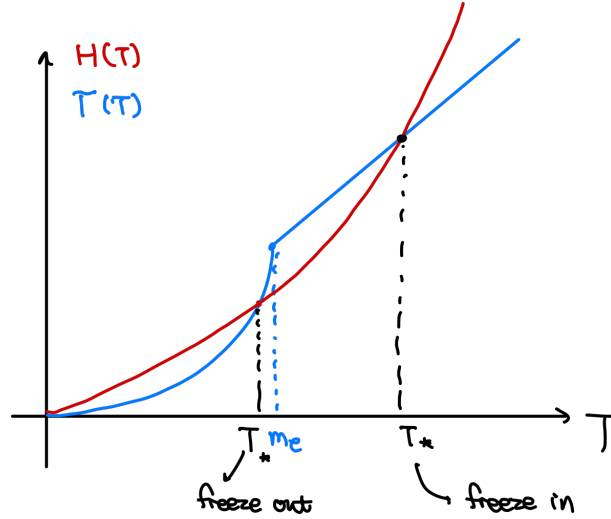


Figure 6: The plot of the rate of reaction Γ and the Hubble parameter H as a function of temperature. The intersection point of the two curves gives us the freeze-out/in temperature of the particle.

we can calculate the freeze-out/in temperature by solving the equation $\Gamma(T_*) = H(T_*)$:

- **freeze in** : this is easy to solve, we have:

$$T_{\text{in}} \sim \alpha^2 M_0 = 10^{14} \text{ GeV} \quad (129)$$

- **freeze out**: this is hard to solve, we can only get an approximate solution by taking the logarithm of the equation, we have:

$$r_e^2 (T m_e)^{\frac{3}{2}} \exp\left(\frac{-m_e}{T}\right) \sim \frac{T^2}{M_0} \quad (130)$$

which gives us:

$$T_{\text{out}} \sim \frac{m_e}{40} = 10 \text{ keV} \quad (131)$$

5.3.3 Physical Interpretation

In this process, in a radiation dominant universe, we assume that **the photon is in thermal equilibrium with the universe**, thus we have $N_\gamma \sim a^{-3}$ only changes due to the expansion of the universe. Then we can use $\frac{N_e}{N_\gamma}$ to represent the number of electrons.

- When $m_e < T < T_{\text{in}}$ the electrons are in thermal equilibrium due to the fact that $\Gamma \gg H$, while we can approximate it as a relativistic particle. Due to the fact that Equation 122, we have:

$$\frac{N_e}{N_\gamma} \sim \frac{g_e}{g_\gamma} \sim 1 \quad (132)$$

- When $T_{\text{out}} < T < m_e$ the electrons are not in thermal equilibrium due to the fact that $\Gamma \ll H$, while we can approximate it as a non-relativistic particle. Due to the fact that the number density of non-relativistic particle is given by Equation 125.

The electron number density shinks exponentially as the universe cools down.

- When $T < T_{\text{out}}$ the scatter is minor and the number of electrons is only affected by the expansion of the universe, thus now we have the number density of electrons given by as $N_e \sim N_* a^{-3}$. where N_* is the number density of electrons at the freeze-out temperature T_{out} .

If we plot the number density of electrons as a function of temperature, we can see that the number density of electrons shinks exponentially as the universe cools down as $T < m_e$, and finally freezes out at T_{out} :

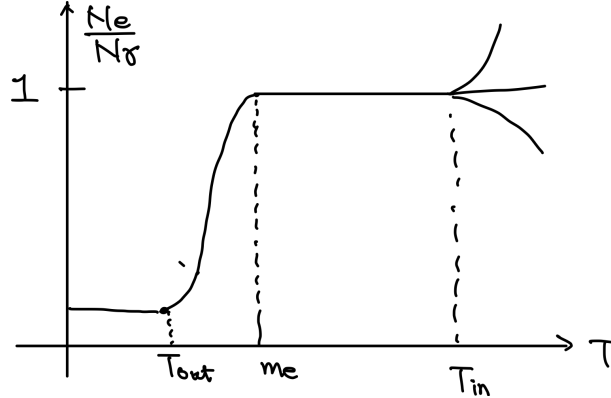


Figure 7: The plot of the number density of electrons as a function of temperature.

An important remark is that the plot is independent of the initial number density and things happening at $T > T_{\text{in}}$.

6 Lecture 6:

6.1 Recombination

Bibliography